Adaptive LQR-Control Design and Friction Compensation for Flexible High-Speed Rack Feeders

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Rack feeders for the automated operation of high bay rackings are of high practical importance. They are characterized by a horizontally movable carriage supporting a tall and flexible vertical beam structure, on which a cage containing the payload can be positioned in vertical direction. To shorten the transport times by using trajectories with increased maximum acceleration and jerk values, accompanying control measures can be introduced counteracting or avoiding undesired vibrations of the flexible structure. In this contribution, both the control-oriented modeling for an experimental setup of such a flexible rack feeder and the model-based design of a gain-scheduled feedforward and feedback control structure are presented. Whereas, a kinematical model is sufficient for the vertical axis, the horizontal motion of the rack feeder is modeled as a planar elastic multibody system with the cage position as scheduling parameter. For the mathematical description of the bending deflections, a one-dimensional Ritz ansatz is introduced. The tracking control design is performed separately for both the horizontal and the vertical axes using decentralized state-space representations. Remaining model uncertainties are estimated by a disturbance observer. The resulting tracking accuracy of the proposed control concept is shown by measurement results from the experimental setup. Furthermore, these results are compared to those obtained with an alternative control concept from previous work. [DOI: 10.1115/1.4025351]

1 Introduction

For the operation of automated high bay rackings, rack feeders with a vertical beam structure are typically used as handling system. Aiming at shorter transport times to further increase the handling capacity, additional control measures can be introduced to counteract structural oscillations of the beam structure, see also Ref. [1]. One possible solution is given by flatness-based feedforward control, where the desired control inputs are determined by dynamic system inversion using the desired trajectories for the flat outputs as described in Refs. [2] and [3]. However, both publications consider only a constant mass position in vertical direction on an elastic beam without any feedback control. A variational approach is presented in Ref. [4] to determine an optimal feedforward control for an elastic beam. The main drawback is that feedforward control alone is not sufficient to guarantee small tracking errors when model uncertainty is present or disturbances act on the system. For these reasons, in the contribution a decentralized gain-scheduled feedforward and feedback control design is presented for fast trajectory control, and an observer-based disturbance compensation is introduced. In contrast to previous work, see Refs. [5] and [6], in this contribution the drive force is employed as control input for the horizontal axis, which corresponds to the use of an underlying control of the motor current on the current converter. A similar configuration is used in Ref. [7], where a passivity-based feedback controller has been designed.

For the experimental investigation of modern control approaches to active oscillation damping as well as tracking control, a test rig of a high-speed rack feeder has been constructed at the Chair of Mechatronics at the University of Rostock, see Fig. 1. The experimental setup consists of a carriage driven by an electric DC servo motor via a toothed belt, on which an elastic beam as the vertical beam structure is mounted. On this beam structure, a cage with the load mass is guided relocatably in vertical direction. This cage with the coordinate $y_K(t)$ in horizontal direction and $x_K(t)$ in vertical direction represents the tool center point (TCP) of the rack feeder that should track desired trajectories as accurate as possible. The movable cage is driven by a tooth belt and an electric DC servo motor as well. The angles of the actuators are measured by internal angular transducers, respectively. Additionally, the horizontal position of the carriage is detected by a magnetostrictive transducer. Two strain gauges are available to determine the bending deformation of the elastic beam. Both axes can be operated with either an underlying velocity control or an underlying current control—directly implemented on the current converter—where the latter corresponds to a force control, as current and force are proportional. In this paper, a tracking controller is designed for the horizontal axis based on the underlying current control, leading to a force input, whereas the vertical axis is operated with an underlying velocity control, i.e., a velocity input. Basis of the control design for the horizontal axis of the rack feeder is a planar elastic multibody system, presented in Sec. 2, where for the mathematical description of the bending deflection of the elastic beam a Ritz ansatz is introduced that covers the first bending mode. In Sec. 3, the decentralized feedforward and feedback control design for both axes is performed employing a linearized state-space representation, respectively. Given couplings between both axes are taken into account by gain-scheduling techniques with the normalized vertical cage position as scheduling parameter, see also Ref. [5]. This leads to an adaptation of the whole control structure of the horizontal axis. As presented in Sec. 4, friction acting on the carriage is compensated in a feedforward manner by a static friction model. Remaining model uncertainties as well as external disturbance forces affecting the motion of the carriage are considered by estimating a lumped disturbance force with a reduced-order disturbance observer, see Sec. 5. The performance of the proposed control concept is shown in Sec. 6 by experimental results from the test setup with regard to tracking

Contributed by the Design Engineering Division of ASME for publication in the Journal of Computational and Nonlinear Dynamics. Manuscript received October 29, 2012; final manuscript received August 30, 2013; published online October 14, 2013. Assoc. Editor: Johannes Gerstmayr.

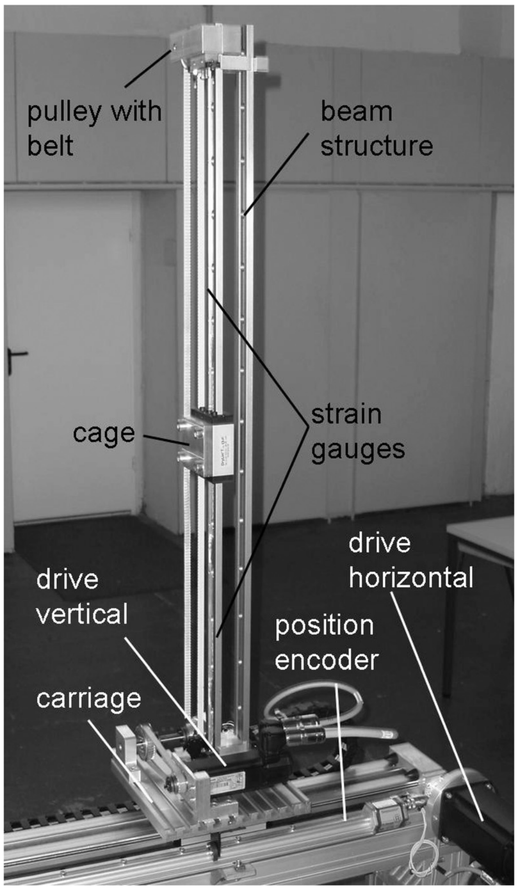


Fig. 1 Experimental setup of the high-speed rack feeder

behavior and damping of bending oscillations. Especially, the artificial damping introduced by the closed control loop represents a main improvement. The maximum velocity of the TCP during the tracking experiments is approx. 2.3 m/s. Finally, a comparison of the tracking accuracy achieved with the proposed control design and corresponding experimental results from previous work [5], with an underlying velocity-control of the carriage, is presented.

2 Modeling of the Mechatronic System

2.1 Control-Oriented Modeling of the Horizontal Axis. Elastic multibody models have proven advantageously for the control-oriented modeling of flexible mechanical systems. For the feedforward and feedback control design of the horizontal axis of the rack feeder, a multibody model with three rigid bodies—the carriage (reduced mass $m_{S,\mathrm{red}}$ including the contribution of the motor inertia), the cage movable on the beam structure (mass m_K, mass moment of inertia θ_K), and the end mass at the tip of the beam (mass m_E)—and an elastic Bernoulli beam (density ρ, cross sectional area A, Young's modulus E, second moment of area I_{zB}, and length l) is chosen, c.f. Fig. 2. The varying vertical position $x_K(t)$ of the cage on the beam is denoted by the dimensionless system parameter

$$\kappa(t) = \frac{x_K(t)}{l} \tag{1}$$

The elastic degrees of freedom of the beam concerning the bending deflection can be described by the following Ritz ansatz

$$v(x,t) = \bar{\bar{v}}_1(x) v_1(t) \tag{2}$$

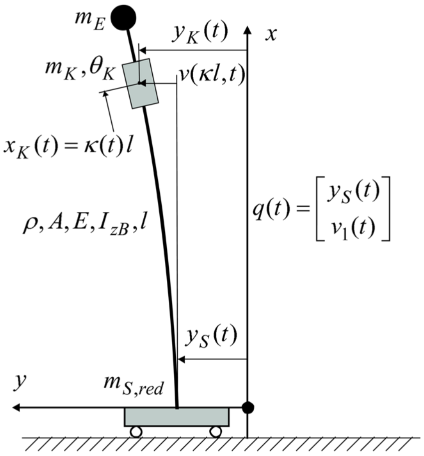


Fig. 2 Elastic multibody model of the rack feeder

with the spatial ansatz function

$$\bar{\bar{v}}_1(x) = \frac{3}{2}\left(\frac{x}{l}\right)^2 - \frac{1}{2}\left(\frac{x}{l}\right)^3 \tag{3}$$

which accounts for the first bending mode. The vector of generalized coordinates results in

$$\mathbf{q}(t) = [\, y_S(t) \; v_1(t)\,]^T \tag{4}$$

The nonlinear equations of motion can be derived either by Lagrange's equations or, advantageously, by the Newton-Euler approach, c.f., Ref. [8].

For this purpose, position vectors to the corresponding centers of gravity are introduced: the position vectors to the carriage $\mathbf{r}_S$, to the cage $\mathbf{r}_K$, to the mass at the beam tip $\mathbf{r}_E$ and to a mass element of the Bernoulli beam $\mathbf{r}_{BE}$ are given by

$$\begin{aligned} \mathbf{r}_S &= \begin{bmatrix} 0 \\ y_S \end{bmatrix}, \quad \mathbf{r}_K = \begin{bmatrix} x_K \\ y_S + v(x_K) \end{bmatrix} \\ \mathbf{r}_E &= \begin{bmatrix} l \\ y_S + v(l) \end{bmatrix}, \quad \mathbf{r}_{BE} = \begin{bmatrix} x_{BE} \\ y_S + v(x_{BE}) \end{bmatrix} \end{aligned} \tag{5}$$

By computing the Jacobian's of translation

$$\mathbf{J}_{Ti} = \frac{\partial \mathbf{r}_i}{\partial \mathbf{q}}, \quad i = \{S, K, E, BE\} \tag{6}$$

for these vectors and the Jacobian's of rotation

$$\mathbf{j}_{Rj} = \frac{\partial \varphi_j}{\partial \mathbf{q}}, \quad j = \{K, E, BE\} \tag{7}$$

for the angles, the nonlinear equations of motion $\tilde{\mathbf{M}}(\mathbf{q})\ddot{\mathbf{q}} + \tilde{\mathbf{k}}(\mathbf{q},\dot{\mathbf{q}}) = \tilde{\mathbf{h}}(\mathbf{q},\dot{\mathbf{q}},u)$ are obtained as follows:

$$\begin{aligned} \tilde{\mathbf{M}} &= m_{S,\mathrm{red}} \mathbf{J}_{TS}^T \mathbf{J}_{TS} + m_K \mathbf{J}_{TK}^T \mathbf{J}_{TK} + m_E \mathbf{J}_{TE}^T \mathbf{J}_{TE} \\ &\quad + \theta_K \mathbf{j}_{RK} \mathbf{j}_{RK}^T + \rho \int_0^l \left(A \mathbf{J}_{\mathrm{TBE}}^T \mathbf{J}_{\mathrm{TBE}} + I_{zB} \mathbf{j}_{\mathrm{RBE}} \mathbf{j}_{\mathrm{RBE}}^T \right) \mathrm{d}x \end{aligned} \tag{8}$$

$$\begin{aligned} \tilde{\mathbf{k}} &= m_{S,\mathrm{red}} \mathbf{J}_{TS}^T \dot{\mathbf{J}}_{TS} \dot{\mathbf{q}} + m_K \mathbf{J}_{TK}^T \dot{\mathbf{J}}_{TK} \dot{\mathbf{q}} + m_E \mathbf{J}_{TE}^T \dot{\mathbf{J}}_{TE} \dot{\mathbf{q}} \\ &\quad + \theta_K \mathbf{j}_{RK} \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{j}_{RK}^T\right) \dot{\mathbf{q}} \\ &\quad + \rho \int_0^l \left(A \mathbf{J}_{\mathrm{TBE}}^T \dot{\mathbf{J}}_{\mathrm{TBE}} \dot{\mathbf{q}} + I_{zB} \mathbf{j}_{\mathrm{RBE}} \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{j}_{\mathrm{RBE}}^T\right) \dot{\mathbf{q}} \right) \mathrm{d}x \end{aligned} \tag{9}$$

$$\tilde{\mathbf{h}} = \mathbf{J}_{TS}^T \begin{bmatrix} F_{SM} - F_{SR} \\ 0 \end{bmatrix} - \frac{\partial U(\mathbf{q})}{\partial \mathbf{q}} - \frac{\partial R(\dot{\mathbf{q}})}{\partial \dot{\mathbf{q}}} \tag{10}$$

with the drive force of the carriage F_{SM} and the associated friction force F_{SR}. Here the potential energy $U(\mathbf{q})$ consists of the gravity potential and the strain energy of the elastic beam. The Rayleigh function $R(\dot{\mathbf{q}})$ allows for an efficient computation of the stiffness-proportional damping matrix. After a linearization for small bending deflections, which holds during the operation of the rack feeder due to model-based trajectory planning according to Sec. 3, the equations of motion can be stated in M-D-K form

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{D}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{h} \cdot [F_{SM}(t) - F_{SR}(\dot{y}_S(t))] \tag{11}$$

The symmetric mass matrix is given by

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{bmatrix} \tag{12}$$

with

$$m_{11} = m_{S,\text{red}} + \rho A l + m_K + m_E \tag{13}$$

$$m_{12} = \frac{3}{8}\rho A l + \frac{m_K \kappa^2}{2}(3-\kappa) + m_E \tag{14}$$

$$m_{22} = \frac{33}{140}\rho A l + \frac{6\rho I_{zB}}{5l} + \frac{m_K \kappa^4}{4}(3-\kappa)^2 + \frac{9\theta_K \kappa^2}{4l^2}(2-\kappa) + m_E \tag{15}$$

The damping matrix, assuming stiffness-proportional damping, and the stiffness matrix become

$$\mathbf{D} = \begin{bmatrix} 0 & 0 \\ 0 & \dfrac{3k_d E I_{zB}}{l^3} \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 0 & 0 \\ 0 & k_{22} \end{bmatrix} \tag{16}$$

with

$$k_{22} = \frac{3EI_{zB}}{l^3} - \frac{3}{8}\rho A g - \frac{3 m_K g \kappa^3}{l}\left(1 + \frac{3\kappa^2}{20} - \frac{3\kappa}{4}\right) - \frac{6 m_E g}{5l} \tag{17}$$

In Eq. (16), the parameter k_d denotes the coefficient of stiffness-proportional damping for the elastic beam. The input vector of the generalized forces, which accounts for the control input as well as the disturbance input, reads $\mathbf{h} = [1\ 0]^T$. The equations of motion can be solved for the second time derivative of the vector of generalized coordinates, which results in

$$\ddot{\mathbf{q}} = -\mathbf{M}^{-1}\mathbf{K}\mathbf{q} - \mathbf{M}^{-1}\mathbf{D}\dot{\mathbf{q}} + \mathbf{M}^{-1}\mathbf{h}F_{SM} - \mathbf{M}^{-1}\mathbf{h}F_{SR} \tag{18}$$

As described in Secs. 4 and 5, the friction force F_{SR} is compensated by a combination of feedforward control and observer-based disturbance compensation. For feedforward and feedback control design; hence, the disturbance force F_{SR} can be neglected. The state-space representation of the horizontal axis—depending on the normalized cage position κ as scheduling parameter—becomes

$$\underbrace{\begin{bmatrix} \dot{\mathbf{q}} \\ \ddot{\mathbf{q}} \end{bmatrix}}_{\dot{\mathbf{x}}_y} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}}_{\mathbf{A}_y(\kappa)} \underbrace{\begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}}_{\mathbf{x}_y} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{h} \end{bmatrix}}_{\mathbf{b}_y(\kappa)} \underbrace{F_{SM}}_{u_y} \tag{19}$$

Note that the actuator dynamics concerning F_{SM} is negligible due to the high bandwidth of the underlying current control.

2.2 Control-Oriented Modeling of the Vertical Axis. The vertical movement of the cage, where an underlying velocity control is employed on the current converter, can be described by a first-order lag system

$$T_{1x}\ddot{x}_K(t) + \dot{x}_K(t) = v_K(t) \tag{20}$$

As the elasticity of the toothed belt for the positioning of the cage is negligible, Eq. (20) represents a kinematical model for the vertical axis and replaces the equation of motion. With the state vector $\mathbf{x}_x = [x_K\ \dot{x}_K]^T$, the state-space description of Eq. (20) follows immediately in the form

$$\dot{\mathbf{x}}_x = \begin{bmatrix} \dot{x}_K \\ \ddot{x}_K \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & -\dfrac{1}{T_{1x}} \end{bmatrix}}_{\mathbf{A}_x} \underbrace{\begin{bmatrix} x_K \\ \dot{x}_K \end{bmatrix}}_{\mathbf{x}_x} + \underbrace{\begin{bmatrix} 0 \\ \dfrac{1}{T_{1x}} \end{bmatrix}}_{\mathbf{b}_x} \underbrace{v_K}_{u_x} \tag{21}$$

The state-space description of the x-axis—using an underlying velocity control—is invariant, whereas the state-space representation of the horizontal y-axis—based on a force input corresponding to an underlying current control—depends on the varying system parameter $\kappa(t)$. A gain-scheduling; hence, is necessary only for the horizontal axis.

3 Decentralized Control Design

3.1 Adaptive Control Design and Friction Compensation. In the decentralized control approach, the coupling of the vertical cage motion with the horizontal axis is taken into account by gain-scheduling techniques. The design of the state feedback for the horizontal motion is carried out on the basis of the linear-quadratic regulator (LQR) approach, where the vector of feedback gains is determined by minimization of a quadratic cost function with a combined weighting of the state variables as well as the control inputs. The control gains follow from a numerical solution of the corresponding algebraic Riccati equation (ARE) in a chosen number of operating points. Due to the dependency of the system matrices on the varying system parameter κ, the ARE becomes also a function of this parameter. To guarantee the continuity of the feedback gains concerning this varying system parameter, constant weightings are employed for the states as well as the control inputs. Starting point for the LQR design is the time-weighted cost function

$$J = \frac{1}{2}\int_0^\infty \left[\mathbf{x}_y^T \mathbf{Q}_y \mathbf{x}_y + r_y u_y^2\right] e^{2\alpha_y t} dt \tag{22}$$

which contains the weighting matrix $\mathbf{Q}_y$, the weight r_y, and an additional parameter α_y. The weighting matrix $\mathbf{Q}_y$ for the state vector $\mathbf{x}_y$ is chosen as a constant, positive definite diagonal matrix

$$\mathbf{Q}_y = \text{diag}[1e4, 8e3, 300, 10] = \text{const.} > 0 \tag{23}$$

the scalar input weight as constant positive value $r_y = 3e-4$. With the parameter $\alpha_y = 4$ an absolute stability margin of the closed-loop eigenvalues can be specified

$$\max_i\{\text{Re}(s_i)\} \leq -\alpha_y = -4 \tag{24}$$

In the s-plane; hence, all closed-loop eigenvalues are located left to a parallel line in the distance α_y from the imaginary axis. In a chosen operating point, characterized by a parameter $p = \kappa$, the optimal feedback control law can be determined with the positive definite solution $\mathbf{P}(\kappa)$ of the parameter-dependent ARE

$$\mathbf{A}_{y\alpha}^T(\kappa)\mathbf{P}(\kappa) + \mathbf{P}(\kappa)\mathbf{A}_{y\alpha}(\kappa) - r_y^{-1}\mathbf{P}(\kappa)\mathbf{b}_y(\kappa)\mathbf{b}_y^T(\kappa)\mathbf{P}(\kappa) + \mathbf{Q}_y = 0 \tag{25}$$

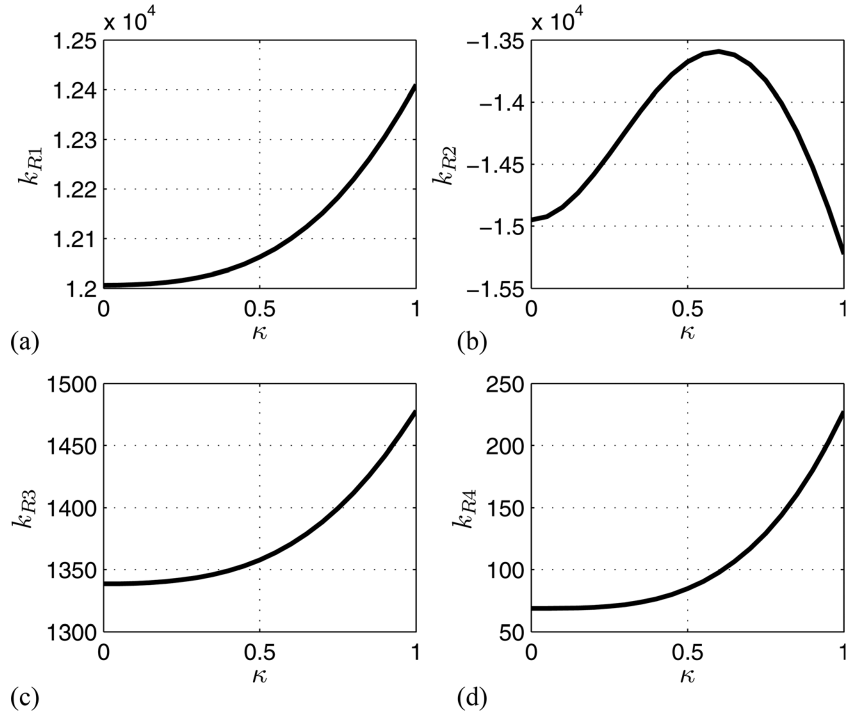


Fig. 3 Feedback control gains—normalized with their corresponding SI-units—as function of the dimensionless vertical position κ as scheduling parameter

The parameter-dependent state feedback becomes

$$u_{y,ZR} = -\mathbf{k}_y^T(\kappa)\mathbf{x}_y \tag{26}$$

with the vector of feedback gains

$$\mathbf{k}_y^T(\kappa) = r_y^{-1}\mathbf{b}_y^T(\kappa)\mathbf{P}(\kappa) \tag{27}$$

The matrix $\mathbf{A}_{y\alpha}$ used in the ARE is derived from the system matrix $\mathbf{A}_y$ according to

$$\mathbf{A}_{y\alpha}(\kappa) = \mathbf{A}_y(\kappa) + \alpha_y\mathbf{I} \tag{28}$$

As the matrix $\mathbf{A}_{y\alpha}(\kappa)$ and the vector $\mathbf{b}_y(\kappa)$ continuously depend on the varying system parameter κ, and constant weightings $\mathbf{Q}_y$ and r_y are employed, the solutions $\mathbf{P}(\kappa)$ of the ARE are continuous functions of κ, as well. The same applies to the vector of feedback gains. Consequently, a gain-scheduled feedback control can be derived by choosing a finite number of operating points in the space of the varying system parameter κ and by performing LQR design with constant weightings.

The resulting feedback gains contained in the vector $\mathbf{k}_y^T$ could be either approximated by simple ansatz functions, e.g., polynomials, or could be interpolated. The simplest way of implementation is given by using lookup tables with a linear interpolation between the chosen operating points, see Fig. 3. These lookup tables are generated automatically and can be integrated in the control structure without any effort. The approximation quality can be easily specified by the number of operating points used for the control design. Obviously, a trade-off has to be found between the desired approximation quality and the necessary storage.

The resulting location of the closed-loop eigenvalues in the s-plane reflecting the operating points used for the design is depicted in Fig. 4. Note that the eigenvalue location characterizes only the special case, where the normalized cage position κ is constant. The asymptotic stability of the time-varying system with the closed-loop system matrix $\mathbf{A}_{yR}(\kappa) = \mathbf{A}_{yR}(\kappa) - \mathbf{b}_y(\kappa)\mathbf{k}_y^T(\kappa)$ can be shown with a quadratic Lyapunov function

$$V(\mathbf{x}_y) = \mathbf{x}_y^T\mathbf{P}\,\mathbf{x}_y \tag{29}$$

where $\mathbf{P}$ represents a matrix that is determined as the symmetric and positive definite solution of the Lyapunov equation

$$\bar{\mathbf{A}}_{yR}^T(\kappa)\mathbf{P} + \mathbf{P}\bar{\mathbf{A}}_{yR}(\kappa) = -\mathbf{Q} \tag{30}$$

with a selected constant matrix $\bar{\mathbf{A}}_{yR} = \bar{\mathbf{A}}_{yR}(\bar{\kappa} = 0.1)$ and the choice $\mathbf{Q} = \mathbf{I}$. A sufficient condition for asymptotic stability is the negative definiteness of the state-dependent matrix

$$\mathbf{P}\mathbf{A}_{yR}(\kappa) + \mathbf{A}_{yR}^T(\kappa)\mathbf{P} < 0 \tag{31}$$

Following this approach with $\bar{\kappa} = 0.1$, the Lyapunov function resulting from Eq. (30) proves the closed-loop stability within the interval $\kappa \in (0, 0.55)$ but also reflects the conservatism due to the arbitrary choice $\mathbf{Q} = \mathbf{I}$. A stability proof for the whole operation range becomes possible by solving a LMI feasibility problem.

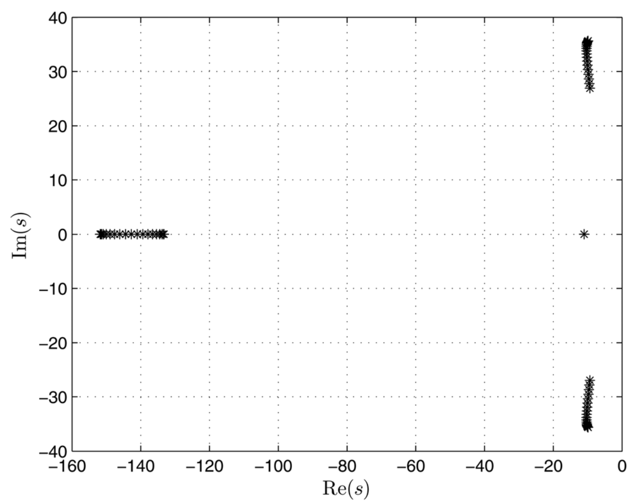


Fig. 4 Closed-loop eigenvalues in dependency on the dimensionless scheduling parameter κ

Using the software libraries YALMIP, see Ref. [9], in combination with SeDuMi [10], a numerical solution $\mathbf{P}$ has been determined that represents a common Lyapunov function for the time-varying system and proves closed-loop stability.

The output equation corresponding to the horizontal cage position as controlled output becomes

$$y_K(t) = \underbrace{\begin{bmatrix} 1 & \frac{1}{2}\kappa^2(3-\kappa) & 0 & 0 \end{bmatrix}}_{\mathbf{c}_y^T} \mathbf{x}_y(t) \tag{32}$$

The command transfer function can be derived as

$$\begin{aligned} Y_K(s) &= \mathbf{c}_y^T \left(s\mathbf{I} - \mathbf{A}_y + \mathbf{b}_y \mathbf{k}_y^T \right)^{-1} \mathbf{b}_y U_V(s) \\ &= \frac{b_0 + b_1 \cdot s + b_2 s^2}{N(s)} \cdot U_V(s) \end{aligned} \tag{33}$$

Obviously, the numerator of the control transfer function contains a second degree polynomial in s, leading to two transfer zeros. This shows that the considered output represents a nonflat output variable that makes feedforward control design more difficult. As the horizontal axis is a fully controllable linear system, which can be easily shown by computing the controllability matrix

$$\mathbf{Q}_{yS} = [\mathbf{b}_y \ \mathbf{A}_y\mathbf{b}_y \ \mathbf{A}_y^2\mathbf{b}_y \ \mathbf{A}_y^3\mathbf{b}_y] \tag{34}$$

and by checking Kalman's rank condition $\text{rank}(\mathbf{Q}_{yS}) = n = 4$, the flat system output could be computed directly

$$y_{y,fl} = a \cdot [0 \ 0 \ 0 \ 1]^T \mathbf{Q}_{yS}^{-1} \mathbf{x}_y \tag{35}$$

Unfortunately, this flat output variable does not comply with the horizontal cage position as controlled variable. Therefore, an alternative way is chosen to derive the feedforward control law. The main idea is given by a modification of the numerator of the control transfer function by introducing a polynomial ansatz for the feedforward control action according to

$$U_V(s) = \left[k_{V0} + k_{V1} \cdot s + \cdots + k_{V4} \cdot s^4 \right] Y_{Kd}(s) \tag{36}$$

For its realization the desired trajectory $y_{Kd}(t)$ as well as the first four time derivatives are available from a trajectory planning module. The feedforward gains can be computed from a comparison of the corresponding coefficients in the numerator as well as the denominator polynomials of

$$\begin{aligned} \frac{Y_K(s)}{Y_{Kd}(s)} &= \frac{(b_0 + b_1 \cdot s + b_2 \cdot s^2)[k_{V0} + \cdots + k_{V4} \cdot s^4]}{N(s)} \\ &= \frac{b_{V0}(k_{Vj}) + b_{V1}(k_{Vj}) \cdot s + \cdots + b_{V6}(k_{Vj}) \cdot s^6}{a_0 + a_1 \cdot s + \cdots + s^4} \end{aligned} \tag{37}$$

according to

$$a_i = b_{Vi}(k_{Vj}), \quad i = 0, \ldots, 4 \tag{38}$$

This leads to parameter-dependent feedforward gains $k_{Vj} = k_{Vj}(\kappa)$. It becomes obvious that due to the higher numerator degree in the modified control transfer function a remaining dynamics must be accepted. Though perfect tracking could not be achieved due to the transfer zeros of the open-loop system, this easily implementable feedforward control contributes significantly to an improved tracking behavior.

3.2 Control Design for the Vertical Axis. For the control of the cage position $x_K(t)$ a simple proportional feedback in combination with feedforward control, which is based on the inverse transfer function of this axis, is sufficient. The control law is chosen as

$$v_K(t) = K_R(x_{Kd}(t) - x_K(t)) + \dot{x}_{Kd}(t) + T_{1x}\ddot{x}_{Kd}(t) \tag{39}$$

The desired trajectory $x_{Kd}(t)$ and its first two time derivatives are provided by a trajectory planning module, c.f. Fig. 6. This results in an asymptotically stable tracking error dynamics

$$T_{1x}\ddot{e}_x(t) + \dot{e}_x(t) + K_R e_x(t) = 0 \tag{40}$$

Note that additional damping could be easily introduced by extending the control law, Eq. (40), by a differential feedback control action.

4 Feedforward Friction Compensation

The main part of the nonlinear friction force $F_{SR}(\dot{y}_S)$ acting on the carriage is counteracted by an experimentally identified static friction model in a feedforward manner. This feedforward friction model $F_{SR,FF}$ depends on the desired carriage velocity $\dot{y}_{Sd}$ and covers Coulomb friction as well as viscous friction

$$F_{SR,FF} = F_C \text{sign}(\dot{y}_{Sd}) + (b_0 - |\dot{y}_{Sd}|b_1)\dot{y}_{Sd} \tag{41}$$

Here, the parameter F_C characterizes Coulomb friction. The ansatz term for the viscous friction is described by the parameters b_0 and b_1. These three parameters of the friction model have been identified by experiments in combination with least-squares techniques. A comparison of the measured friction force and the identified one is depicted in Fig. 5. It shows that the simulation with the identified friction characteristic matches well the measured friction force. To avoid switchings at zero velocity $\dot{y}_{Sd} = 0$, the function $\text{sign}(\dot{y}_{Sd})$ can be approximated by the smooth function $\tanh(\dot{y}_{Sd}/\varepsilon)$, with $\varepsilon > 0$. The desired output variable and a certain number of its time derivatives is available from trajectory planning. These can be used to calculate the desired system states. The desired carriage velocity $\dot{y}_{Sd}$, for instance, can be derived from the available inverse system model in the Laplace domain. For this purpose, the polynomial feedforward law $U_V(s)$ has to be partially differentiated with respect to the corresponding feedback gain. The desired carriage velocity $\dot{y}_{Sd}$ results in

$$\dot{y}_{Sd}(s) = \left[\frac{\partial k_{V0}}{\partial k_{R3}} + \frac{\partial k_{V1}}{\partial k_{R3}} \cdot s + \cdots + \frac{\partial k_{V4}}{\partial k_{R3}} \cdot s^4 \right] Y_{Kd}(s) \tag{42}$$

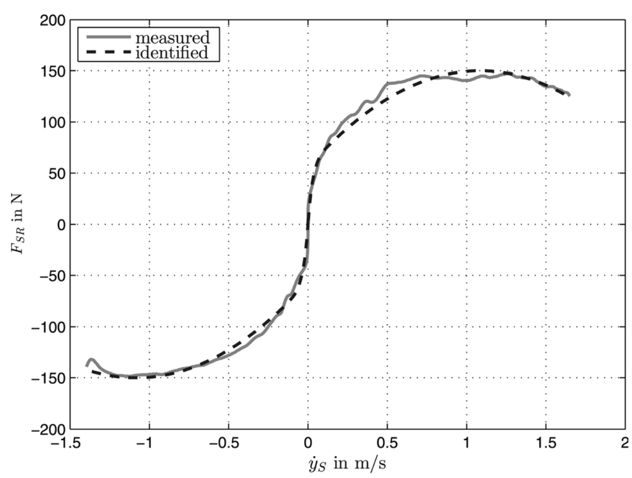


Fig. 5 Comparison of the measured friction force and the experimentally identified friction force

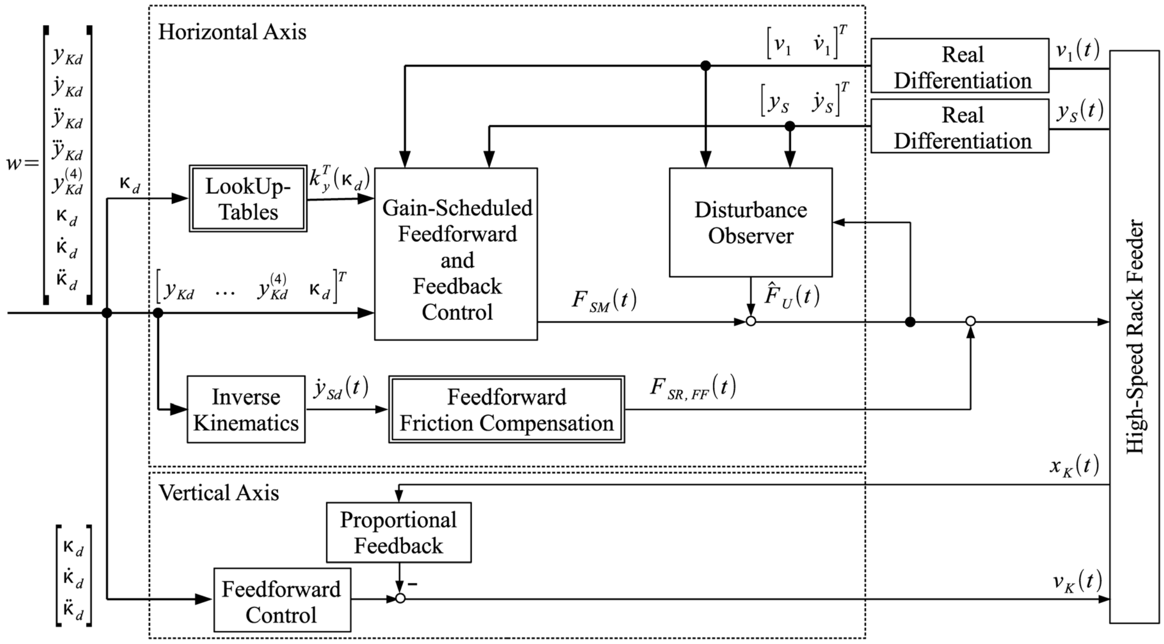


Fig. 6 Implementation of the cascaded control structure

5 Reduced-Order Disturbance Observer

The remaining model uncertainties $F_U(t) = F_{SR}(t) - F_{SR,FF}(t)$, which are not compensated by the feedforward friction model, are estimated by a reduced-order disturbance observer as described in Ref. [11]. Please note that this lumped disturbance model does not only take into account imperfections of the static friction model but also all unmodeled forces affecting the motion of the carriage. The observer design is based on the equations of motion. The key idea for the observer design is to extend the state equations with an integrator as disturbance model

$$\begin{aligned} \dot{\mathbf{x}}_y &= \mathbf{f}_y\left(\mathbf{x}_y, F_U, u\right) \\ \dot{F}_U &= 0 \end{aligned} \tag{43}$$

where $\mathbf{x}_y = [y_S\ v_1\ \dot{y}_S\ \dot{v}_1]^T$ denotes the measurable state vector. The estimated disturbance force $\hat{F}_U$ is obtained from

$$\hat{F}_U = \mathbf{h}_y^T \mathbf{x}_y + z \tag{44}$$

with the chosen observer gain vector

$$\mathbf{h}_y^T = [\,0\ 0\ 0\ h_4\,] \tag{45}$$

The state equation for z is given by

$$\dot{z} = \Phi\left(\mathbf{x}_y, \hat{F}_U, u\right) \tag{46}$$

The observer gain vector $\mathbf{h}_y$ and the vector function Φ have to be chosen such that the steady-state observer error $e = F_U - \hat{F}_U$ converges to zero. Thus, the vector function Φ can be determined as follows

$$\dot{e} = 0 = \dot{\hat{F}}_U - \mathbf{h}_y^T \mathbf{f}_y\left(\mathbf{x}_y, \hat{F}_U, u\right) - \Phi\left(\mathbf{x}_y, \hat{F}_U, u\right) \tag{47}$$

This yields an equation for the calculation of the observer term $\Phi\left(\mathbf{x}_y, \hat{F}_U, u\right)$

$$\Phi\left(\mathbf{x}_y, \hat{F}_U, u\right) = -\mathbf{h}_y^T \mathbf{f}_y\left(\mathbf{x}_y, \hat{F}_U, u\right) \tag{48}$$

The error dynamics $\dot{e}$ has to be asymptotically stable. Accordingly, the eigenvalue of the Jacobian

$$J_e = \frac{\partial \Phi\left(\mathbf{x}_y, F_U, u\right)}{\partial F_U} \tag{49}$$

must be located in the left complex half-plane. This can be achieved by proper choice of the observer gain h_4. The implementation of the overall control structure including the reduced-order observer is illustrated in Fig. 6. Simulations show that this observer-based disturbance compensation leads to a robustification of the control approach.

6 Experimental Results

The benefits and the efficiency of the proposed control structure shall be pointed out by experimental results obtained from the test setup available at the Chair of Mechatronics, University of Rostock.

For this purpose, a synchronous and four times continuously differentiable desired trajectory is considered for the position of the cage in both x- and y-direction. The desired trajectory is given by polynomial functions that comply with specified kinematic constraints, which is achieved by taking advantage of time scaling techniques. The desired trajectory shown in Fig. 7 comprises a sequence of reciprocating motions with maximum velocities of 2 m/s in horizontal direction and 1.0 m/s in vertical direction.

6.1 Experimental Results of the Proposed Control Structure.

The resulting tracking errors of the proposed control structure

$$e_y(t) = y_{Kd}(t) - y_K(t) \tag{50}$$

and

$$e_x(t) = x_{Kd}(t) - x_K(t) \tag{51}$$

are depicted in Fig. 8. As can be seen, the maximum position error during the movements is 3.3 mm in y-direction and about 1.2 mm in x-direction. The steady-state position error in y- as well as in x-direction is smaller than 0.5 mm. Figure 9 shows the comparison of the bending deflection measured by strain gauges attached to the flexible beam with desired values. The desired values of the bending deflection v_{1d} are obtained in the Laplace domain, c.f. Eq. (42), as follows

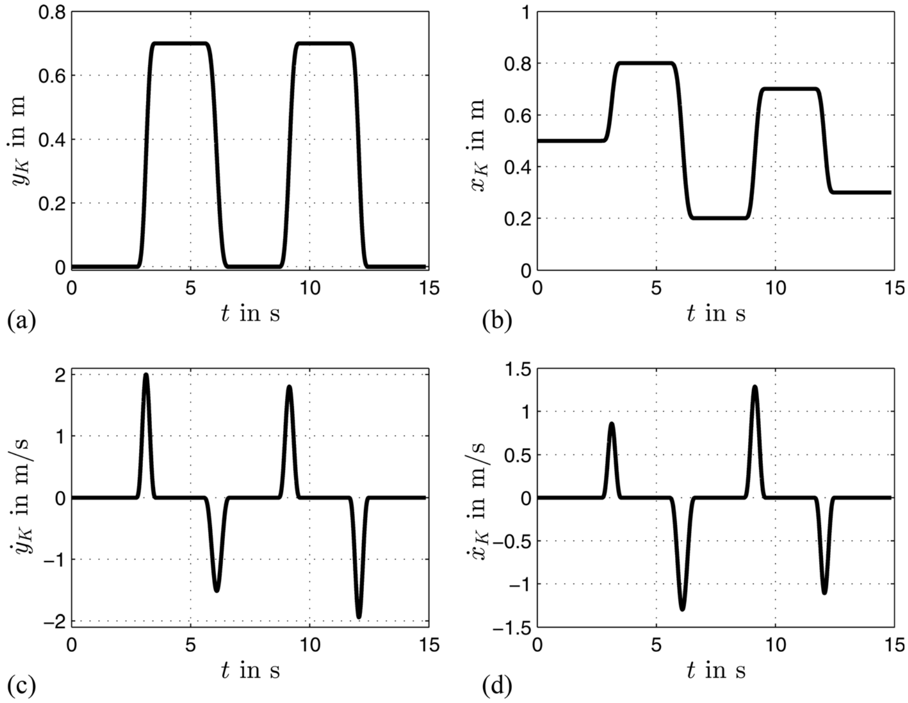


Fig. 7 Desired trajectories for the cage motion: desired position in horizontal direction (*a*), desired position in vertical direction (*c*), desired velocity in horizontal direction (*b*) and desired velocity in vertical direction (*d*)

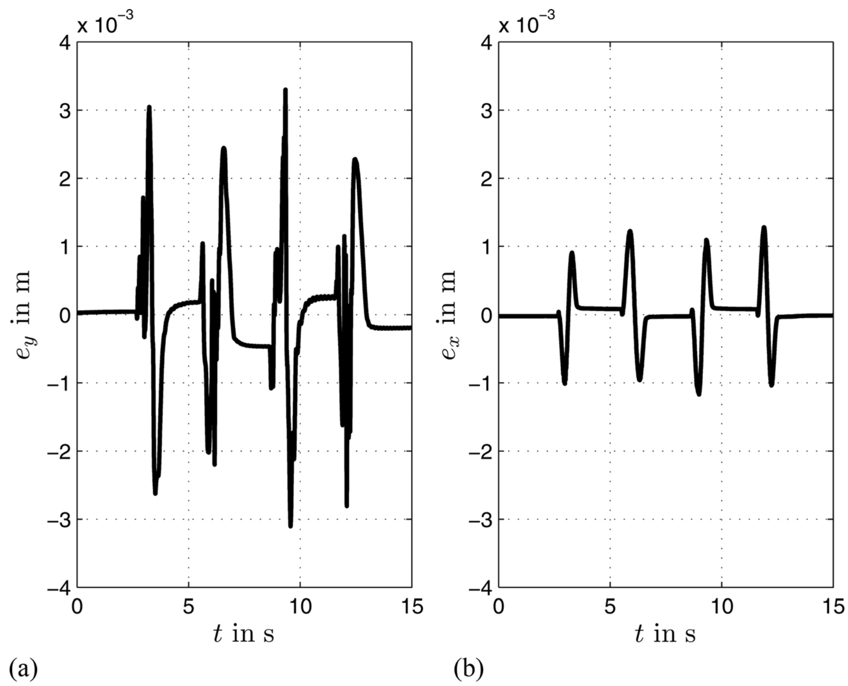


Fig. 8 Tracking error e_y for the cage motion in horizontal direction (*a*) and tracking error e_x for the cage motion in vertical direction (*b*)

$$v_{1d}(s) = \left[\frac{\partial k_{V0}}{\partial k_{R2}} + \frac{\partial k_{V1}}{\partial k_{R2}} \cdot s + \cdots + \frac{\partial k_{V4}}{\partial k_{R2}} \cdot s^4\right] Y_{Kd}(s) \tag{52}$$

During the acceleration as well as the deceleration intervals, physically unavoidable bending deflections can be noticed. The achieved benefit is given by the fact that the remaining oscillations are negligible when the rack feeder arrives at its target position. The good matching of the measured bending deflection with the desired bending deflection—calculated with the derived control-oriented model—reflects the achieved high model quality and justifies the employed one-dimensional Ritz ansatz, Eq. (2).

Figure 10 depicts the disturbance rejection properties due to an external excitation by hand. At the beginning, the control structure is deactivated, and the excited bending oscillations decay only due to the very low material damping. After approx. 2.5 s, the control structure is activated and; hence, the bending oscillations are actively damped.

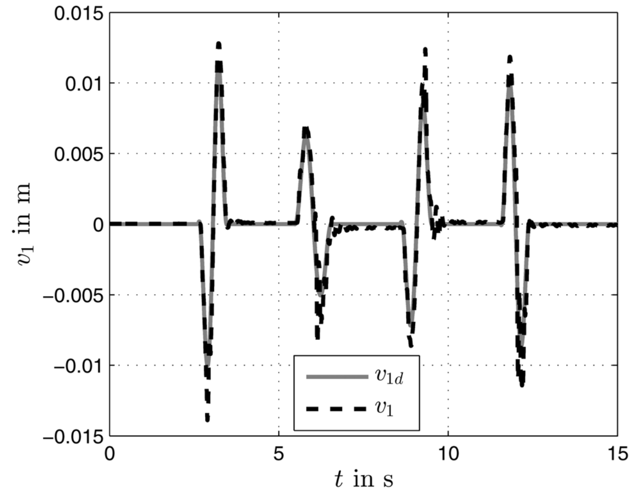


Fig. 9 Comparison of the desired values v_{1d} and the actual values v_1 for the bending deflection

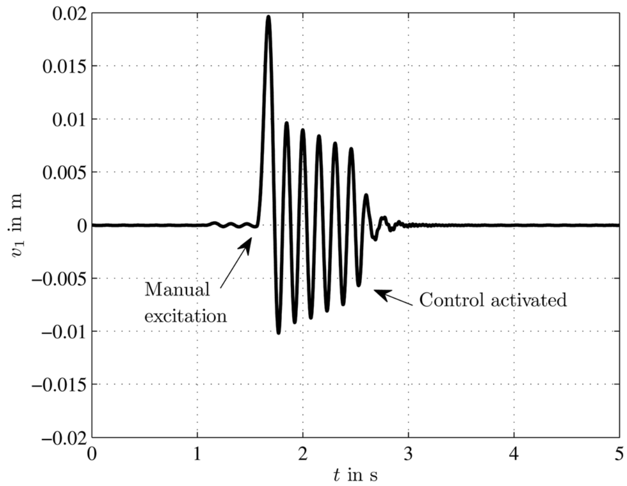


Fig. 10 Transient response after a manual excitation of the bending deflection: at first without feedback control, after approx. 2.5 s with active control

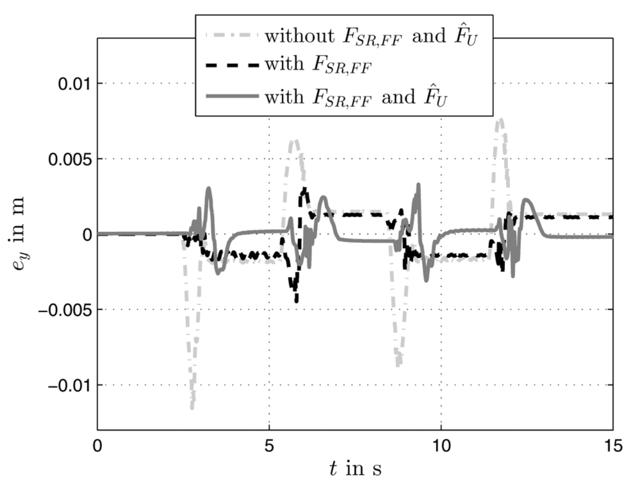


Fig. 11 Comparison of the tracking error e_y with and without considering the disturbance compensation by the feedforward friction compensation part $F_{SR,FF}$ and the estimated lumped disturbance force $\hat{F}_U$

Table 1 Comparison of tracking errors without and with compensating control actions

	CS1	CS2	CS3
$e_{y,\,\mathrm{RMS}}$ in mm	2.45	1.17	0.79
$\lvert e_{y,\,\mathrm{RMS}}\rvert$ in mm	7.73	3.26	3.30

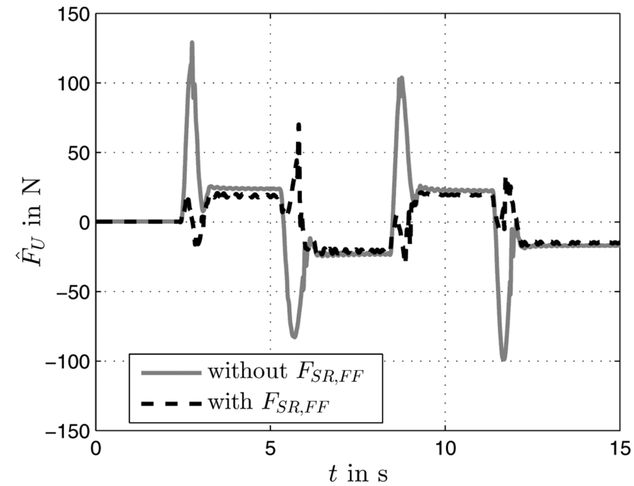


Fig. 12 Comparison of the estimated disturbance force $\hat{F}_U$ with and without employing the feedforward friction compensation $F_{SR,FF}$

Table 2 Comparison of the estimated disturbance force with and without feedforward friction compensation $F_{SR,FF}$

	Without $F_{SR,FF}$	With $F_{SR,FF}$
$\hat{F}_{U,\mathrm{RMS}}$ in N	29.7	15.3
$\lvert\hat{F}_U\rvert$ in N	129.1	70.1

Table 3 Comparison of tracking error measures

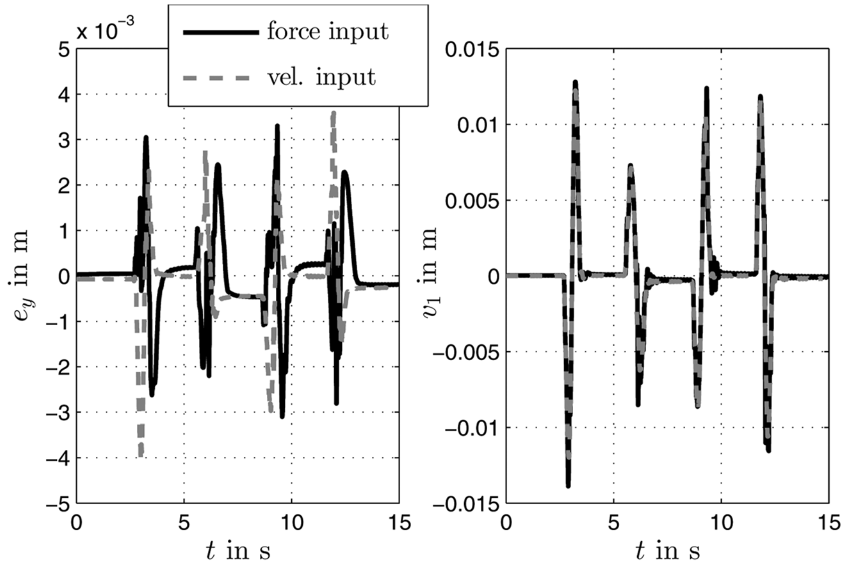
	Force input	Velocity input
$e_{y,\,\mathrm{RMS}}$ in mm	0.79	0.78
$\lvert e_{y,\,\max}\rvert$ in mm	3.30	3.59

The influence of the static friction model and the disturbance observer are shown in Fig. 11 to assess the efficiency of the compensating control actions: (1) without $F_{SR,FF}$ and without $\hat{F}_U$ (CS1), (2) with $F_{SR,FF}$ but without $\hat{F}_U$ (CS2), and (3) with both $F_{SR,FF}$ and $\hat{F}_U$ (CS3). For a comparison of the three different compensation strategies the root-mean square error of the tracking error

$$e_{y,\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(e_y(i)\right)^2} \tag{53}$$

as well as the maximum absolute error $\lvert e_{y,\max}\rvert$ are stated in Table 1.

By introducing the feedforward friction compensation $F_{SR,FF}$ (CS2), the tracking error e_y is reduced significantly in comparison to the case without any friction compensation (CS1). An additional compensation of the remaining disturbances by means of the reduced-order disturbance observer (CS3) leads to a further improvement regarding the tracking behavior, where especially the steady-state error is smaller than in the cases without the observer-based disturbance compensation (CS1 and CS2).



(a) (b)

Fig. 13 Comparison of the tracking error e_y (*a*) and the bending deflection v_1 (*b*) for employing either the drive force or the carriage velocity as input variable

The benefits of the feedforward friction compensation $F_{SR,FF}$ are also shown in Fig. 12. By employing the static friction model for a feedforward friction compensation, the estimated disturbances are reduced significantly, see also the uncertainty measures stated in Table 2.

6.2 Comparison With Experimental Results Based on a Velocity Input. In the following, the experimental results for the horizontal axis are compared to those obtained in previous work [5], with the same control approach but by employing a velocity input. In Fig. 13(*a*), the tracking error of the carriage position in horizontal direction e_y of the proposed control structure—using the force input—is compared to the corresponding results using a velocity input.

Both control approaches yield similar results. Whereas the RMS-error is almost identical, the use of the force input slightly reduces the maximum absolute tracking error $|e_{y,\,\max}|$, see Table 3. This can be explained by the direct compensation of nonlinear friction as well as remaining model uncertainty. The usage of an underlying controller for the carriage velocity like in Ref. [5], however, results in a reduced implementation effort. Particularly, neither friction compensation nor disturbance observer are needed since the friction acting on the carriage is compensated indirectly by the fast underlying velocity control.

Furthermore, the measured bending deflection $v_1(t)$ is compared for both cases in Fig. 13(*b*). The good compliance achieved serves as a validation of the derived control-oriented model.

7 Conclusions and Outlook

In this paper, a gain-scheduled control strategy for flexible high-speed rack feeders in light-weight construction is presented. The decentralized control design is based on a control-oriented elastic multibody system for the horizontal axis with the carriage force as control input, and on a kinematic model for the vertical axis with the velocity as control input. Beneath an active oscillation damping of the first bending mode, an accurate trajectory tracking for the cage position in x- and y-direction is achieved by feedback control in combination with feedforward control. Nonlinear friction is compensated directly in a feedforward manner using an identified static friction model. The remaining uncertainties are estimated by a reduced-order disturbance observer and used for a disturbance compensation. Experimental results from a prototypic test setup point out the benefits of the proposed control structure. Experimental results show maximum tracking errors of 3.3 mm in transient phases, whereas the steady-state tracking error is smaller than 0.5 mm. This represents a slight improvement as compared to previous work, where the same control approach was used for the horizontal axis based on the velocity as control input. Future work will address a model extension towards a consideration of the second bending mode and the design of an observer-based adaptive control. Moreover, possible benefits of a friction compensation based on dynamical friction models shall be considered.

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